

Network Reconnaissance & Port Scanning Audit

Authorized Simulated Target Environment

Prepared by: Darshan Suresh
Date: 27 September 2026
Tools: Nmap, Wireshark

1. Executive Summary

This report documents an authorized network reconnaissance exercise performed against a simulated/sandbox environment. The assessment covers TCP SYN scanning, UDP discovery, service/version identification, operating-system detection, and packet-level analysis using Wireshark.

Actual target-specific results must be inserted from the evidence generated during the lab. No production or third-party systems should be scanned.

2. Scope & Methodology

The methodology uses progressively detailed Nmap scans: TCP SYN discovery, UDP top-port discovery, service/version detection, OS detection, and a combined audit scan. Wireshark is then used to inspect protocol behavior and identify insecure or anomalous traffic.

3. Nmap Findings

Port	Protocol	State	Service	Version	Notes
[insert]	[TCP/UDP]	[open]	[insert]	[insert]	[insert]
[insert]	[TCP/UDP]	[open]	[insert]	[insert]	[insert]
[insert]	[TCP/UDP]	[filtered]	[insert]	[insert]	[insert]

4. Wireshark Analysis

Record the protocols observed, relevant packet numbers, source/destination hosts, and whether traffic is encrypted. If plaintext authentication is observed in the authorized lab, describe the protocol and security implication without publishing real credentials.

Evidence placeholders: 01_capture_overview.png, 02_protocol_hierarchy.png, 03_relevant_stream.png.

5. Potential Attack Vectors

For each exposed service, document the associated risk based on the observed service, version, configuration, and protocol security. Avoid claiming a vulnerability solely from an open port; validate with authoritative service documentation or a controlled vulnerability assessment.

6. Recommendations

Restrict unnecessary services, segment exposed systems, replace plaintext protocols with encrypted alternatives, apply security patches, enforce strong authentication, monitor unusual traffic, and limit administrative services through firewall/ACL controls.

7. Conclusion

The assessment provides a repeatable baseline for understanding the simulated target's network exposure. Final conclusions should be based on the actual Nmap outputs and Wireshark evidence attached to the repository.

Appendix A — Evidence Index

Nmap outputs: evidence/nmap/

Wireshark screenshots: evidence/wireshark/

Execution scripts: scripts/